Training course on Optimal experimental designs
1.Foundations for Linear and nonlinear models

Jesús López Fidalgo
fidalgo@unav.es

mODa14
Model-Oriented Data Analysis and Optimum Design

Drongen 2026

Outline

- Motivating ideas.
- Linear Models.
- Optimal Experimental Design.
- Non-linear models.

Motivating ideas

Fisher principles

Randomization: Randomized Controlled Trials (RCTs).

- No just apparent disorder: Same chance to be chosen for each one.
- Eliminates systematic bias: Patients knowing they received the active drug may report feeling better.
- Mitigates confounding: e.g. older patients are more likely to receive Drug A and also more likely to die.
- Cornerstone of causal inference (Rubin, Pearl).

Replication: of the full experiment to help identify the sources of variation ($\neq$ repeated measurements).

Blocking: Grouping experimental units into homogeneous blocks and randomizing within each block.

- Reduces known but irrelevant sources of variation.
- Increases efficiency without increasing sample size.
- E.g.: multicenter trials, crossover designs receiving treatments at random, animal litter...

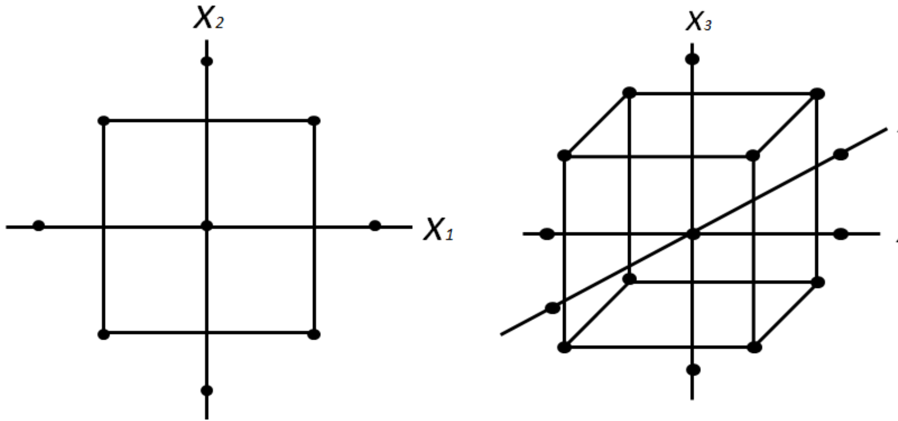
OED natural evolution of Fisher's principles by using information theory: Where and how often to observe!

Montgomery guidelines

- Pre-experimental planning:
  - Recognition and statement of the problem.
  - Model: Choice of the response variable, factors, levels and ranges.
- Choice of experimental design.
- Performing the experiment (monitor the process).
- Statistical analysis of the data.
- Conclusions and recommendations.

Illustrative example: Improving wine fermentation

Central Composite Design (CCD)



Motivating examples: Michaelis-Menten

- Velocity in enzymatic reactions: one substrate with substrate-enzyme complex approximately at steady state.
- $y = \frac{Vx}{x+K} + \epsilon$, where
  V is the maximum rate,
  K is the Michaelis constant,
  $x \in [0, bK]$ the substrate concentration.
- Adding a linear term: $y = \frac{Vx}{x+K} + Fx + \epsilon$
  Example: effect of the isotope transporting glucose in the presence of a mixture of old and young cells,
  e.g. placenta immediately after birth: For young cells a linear model, but for old cells an MM model may fit better.
- Bisubstrate kinetics two-variable model: $y = \frac{Vx_1x_2}{(K_1+x_1)(K_2+x_2)} + \epsilon$.

Non-linear on x's and K's.

Linear Models

All is regression

- $y = \theta^T f(x) + \varepsilon$,
- $\theta^T = (\theta_1, \dots, \theta_m)$ unknown parameters,
- $f^T(x) = (f_1(x), \dots, f_m(x))$ regressors,
- x experimental condition chosen on a *design space*, χ (e.g. $[0, 1]$, $\{0.1, 0.2, 0.5, 0.8, 0.9\}$)
- Response y: normal with mean $\theta^T f(x)$ and constant variance σ^2.
- Independence, normality, homoscedasticity and linearity of the mean.

Examples of linear models

1. Simple linear regression:
   $y = \theta_0 + \theta_1 x + \varepsilon$, $f^T(x) = (1, x)$, $x \in \chi = [a, b]$.
2. Multiple linear regression:
   $y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \varepsilon$, $f^T(x) = (1, x_1, x_2, x_3)$,
   $x \in \chi = [a_1, b_1] \times [a_2, b_2] \times [a_3, b_3]$.
3. Quadratic regression:
   $y = \theta_0 + \theta_1 x + \theta_2 x^2 + \varepsilon$, $f^T(x) = (1, x, x^2)$, $x \in \chi = [a, b]$.
4. Analysis of the Variance (ANOVA) (1-way, 3 levels):

$$
\begin{aligned}
y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \varepsilon,\ f^T(x_1, x_2) = (1, x_1, x_2),\\
x = (x_1, x_2) \in \chi = \{(0, 1), (1, 0), (1, 1)\}.
\end{aligned}
$$

Linear / non-linear regression / model

- All linear:
  - Simple: $y = \theta_0 + \theta_1 x + \varepsilon$.
  - Multiple: $y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \varepsilon$.
- Nonlinear:
  - In the variables (still **linear model**): $y = \theta_0 + \theta_1 x^2 + \varepsilon$.
  - In both parameters and variables (**nonlinear model**): $y = \frac{Vx}{x+K} + Fx + \epsilon$.
- Generalized linear models (e.g. logistic regression with binary response):

$$P(y = 1|x, \theta) = F(\theta_0 + \theta_1 x_1 + \theta_2 x_2).$$

Linear model in matricial form

$Y = (y_1, \ldots, y_n)^T$, uncorrelated observations with constant σ^2.
at experimental conditions $x_1, \ldots, x_n$ (**exact design of size** n).
Objective: MLE of $\theta_1, \ldots, \theta_m, \sigma^2$ and make inferences.

$$X = \begin{pmatrix} f_1(x_1) & \cdots & f_m(x_1) \\ & \cdots & \\ f_1(x_n) & \cdots & f_m(x_n) \end{pmatrix},$$

with the typical notation $x_{ij} = f_j(x_i)$, $i = 1, \ldots, n$; $j = 1, \ldots, m$.
$\mathcal{E} = (\varepsilon_1, \cdots, \varepsilon_n)^T$, actual errors.
Thus $Y = X\theta + \mathcal{E}$, and the MLE (LSE):

$$\hat{\theta} = (X^T X)^{-1} X^T Y, \; \Sigma_{\hat{\theta}} = \sigma^2 (X^T X)^{-1}.$$

Residuals: $E = Y - X\hat{\theta}$.

Optimal Experimental Design

Optimal Experimental Design

- Some of the $x_1, \ldots, x_n$ may be repeated (probability measure): $\xi_n(x) = \frac{n_x}{n}$, where n_x = times x appears in the design.
- Covariance matrix: $\Sigma_{\hat{\theta}} = \sigma^2 (X^T X)^{-1} = \sigma^2 n^{-1} M^{-1}(\xi)$,
- $M(\xi_n) = \sum_{x \in \chi} f(x) f^T(x) \xi_n(x)$ (**information matrix**):
  - Symmetric.
  - Non-negative definite.
  - Singular if the number of different points in the design $k < m$.

Extended concept of experimental design

- **Approximate design (Kiefer):** probability measure with finite support on χ,

$$\xi = \left\{ \begin{array}{cccc} x_1 & x_2 & \dots & x_k \\ p_1 & p_2 & \dots & p_k \end{array} \right\}$$

- In practice, perform $n_i \approx n\xi(x_i)$ experiments at x_i, $\sum_i n_i = n$.
- Information Matrix: $M(\xi) = \sum_{x \in \chi} f(x) f^T(x) \xi(x)$

Rounding off?

$$\xi = \left\{ \begin{array}{ccc} 0 & 0.5 & 1 \\ 0.2 & 0.4 & 0.4 \end{array} \right\}.$$

If just $n = 12$ experiments can be performed, possible rounding off:

	ξ_1	ξ_2	ξ_3	
about $0.2 \times 12 = 2.4$	2	3	3	experiments at 0,
about $0.4 \times 12 = 4.8$	5	4	5	experiments at 0.5,
about $0.4 \times 12 = 4.8$	5	5	4	experiments at 1.

Example

$$y = \theta_0 + \theta_1 x + \varepsilon,\ x \in \chi = [0,1],\ f(x) = (1, x)^T.$$

$$\xi = \left\{ \begin{array}{ccc} 0 & 0.5 & 1 \\ 0.2 & 0.4 & 0.4 \end{array} \right\}.$$

$$M(\xi) \quad = \quad \sum_{i=1}^{3} \left(\begin{array}{cc} 1 & x_i \\ x_i & x_i^2 \end{array} \right) p_i = \left(\begin{array}{cc} 1 & 0.6 \\ 0.6 & 0.5 \end{array} \right)$$

$$\Sigma_{\hat{\theta}} = \frac{\sigma^2}{n} M^{-1}(\xi) = \left(\begin{array}{cc} 3.57 & -4.29 \\ -4.29 & 7.14 \end{array} \right)$$

- $\Phi : \mathcal{M} \to [0, +\infty)$
  to be minimized.
- Non-increasing (better estimates of the parameters).
- It may be global or partial (interest on part of the parameters).
- Positive homogeneous: $\Phi(\delta M) = \frac{1}{\delta}\Phi(M), \quad \delta > 0,$
- A Φ–**optimal design** ξ^* minimizes Φ.
- Convex: $\Phi[(1-\epsilon)\xi + \epsilon\xi'] \leq (1-\epsilon)\Phi(\xi) + \epsilon\Phi(\xi')$.

Equivalence theorem for differentiable criteria

Sensitivity function:

$$\begin{aligned}\psi(x,\xi) &= \partial\Phi[M(\xi), f(x)f^T(x)] \\ &= f^T(x)\nabla\Phi[M(\xi)]f(x) - \mathrm{tr}M(\xi)\nabla\Phi[M(\xi)].\end{aligned}$$

Equivalence theorem:
ξ^* is Φ–optimal if and only if $\psi(x,\xi^*) \geq 0,\ x \in \chi$.
Equality in the support of ξ^*.

D-optimal design maximizes the determinant (minimizes the determinant of the covariance matrix and the volume of the confidence ellipsoid).

Design efficiency of ξ:

$$\left(\frac{\det[M(\xi)]}{\det[M(\xi^*)]}\right)^{1/m}.$$

Alphabetical criteria

D-optimality: $\Phi_D[\xi] = -\log\det M(\xi)$.

G-optimality: $\Phi_G[\xi] = \max_{x\in\chi} f^T(x)M^{-1}(\xi)f(x) \propto \max_{x\in\chi} \mathrm{var}_\xi \hat{y}(x)$.

E-optimality: $\Phi_E[\xi] = \lambda_\xi^{-1}$.

A-optimality: $\Phi_A[\xi] = \mathrm{tr} M^{-1}(\xi) \propto \sum_i \mathrm{var}_\xi \hat{\theta}_i$.

L-optimality: $\Phi_L[\xi] = \mathrm{tr} W M^{-1}(\xi)$, where $W \geq 0$.

I-optimality: $\Phi_I[\xi] = \int_{x\in\chi} f^T(x)M^{-1}(\xi)f(x)\mu(dx)$.

D_s-optimality: $\Phi_{D_s}[\xi] = \det[M^-(\xi)]_s$.

c-optimality: $\Phi_c[\xi] = c^T M^-(\xi)c \propto \mathrm{var}(c^T\hat{\theta})$.

OED for non-linear models

Statistical parametric model

- Model: Family of distributions with pdf's $h(y|x, \theta)$.
- Here is everything: t-tests, ANOVA, regression with correlated or uncorrelated observations or mixed models.

Fisher Information

- The Fisher information matrix (FIM) for a design ξ is then:

$$M(\xi,\theta) = \sum_{x\in\chi} \xi(x) I(x,\theta),$$

where

$$I(x,\theta) = \mathrm{E}_h\left[\left(\frac{\partial L(\theta)}{\partial\theta}\right)\left(\frac{\partial L(\theta)}{\partial\theta^T}\right)\right] = -\mathrm{E}_h\left[\left(\frac{\partial^2 L(\theta)}{\partial\theta^2}\right)\right],$$

and $L(\theta) = \log h(y|x,\theta)$ the log-likelihood.

Fisher Information

- Cramér-Rao Theorem: If $E_\theta(\hat{\theta}) = \omega(\theta)$, then

$$\Sigma_{\hat{\theta}} \geq n^{-1} \frac{\partial \omega(\theta)}{\partial \theta} M^{-1}(\xi, \theta) \left(\frac{\partial \omega(\theta)}{\partial \theta} \right)^T .$$

- The MLE achieves this bound asymptotically.
- If the MLE is unbiased then $\Sigma_{\hat{\theta}} \approx n^{-1} M^{-1}(\xi, \theta)$ asymptotically.

Equivalence theorem for nonlinear models

- $\partial\Phi[M(\xi^*,\theta), M(\xi,\theta)] \geq 0$, for any ξ.
- If differentiable $\psi(x,\xi^*,\theta) \geq 0,\ x \in \chi$, where

$$\begin{aligned}\psi(x,\xi,\theta) &= \partial\Phi[M(\xi,\theta), I(x,\theta)] \\ &= \mathrm{tr}\{[I(x,\theta) - M(\xi,\theta)]\nabla\Phi[M(\xi,\theta)]\}.\end{aligned}$$

Problems

- Dependence on the parameters.
- Enough sample size for convergence.
- When the expectations cannot be computed analytically:

$$\mathsf{E}_\theta\left[\frac{\partial L(\theta)}{\partial \theta}\frac{\partial L(\theta)}{\partial \theta^T}\right] = -\mathsf{E}_\theta\left[\frac{\partial^2 L(\theta)}{\partial \theta^2}\right].$$

- When does there exist $f(x,\theta)$ such that $I(x,\theta) \propto f(x,\theta)f^T(x,\theta)$? For the exponential family (e.g. normality $f(x,\theta)$ is the gradient of the trend).

Michaelis-Menten

- y normally distributed with constant variance and mean:

$$E(y) = \frac{Vx}{K + x} \equiv \eta(x, \theta), x \in [0, bK], \theta^T = (K, V, F).$$

- Equivalent linear model (FIM):

$$f(x) = \left(\frac{\partial \eta}{\partial \theta}\right)_{(K_0, V_0)} = \left(\frac{-V_0 x}{(K_0 + x)^2}, \frac{x}{(K_0 + x)}\right).$$

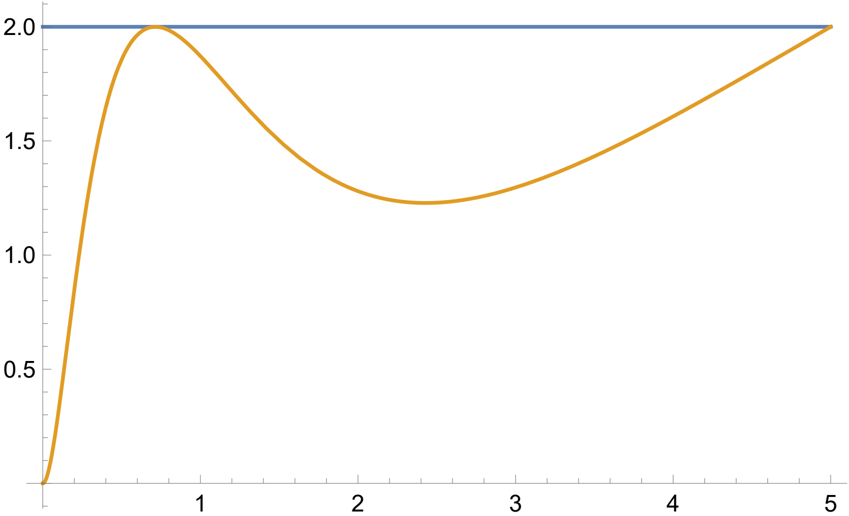
- FIM: $M(\xi) = \sum_x \xi(x) f(x) F^T(x)$.
- Two ways of computing the optimal design:
  1. Assume the optimal design is a 2-point design. Then GET verification.
  2. Graphical method for c-optimality.
  3. Algorithmic computation.

Results

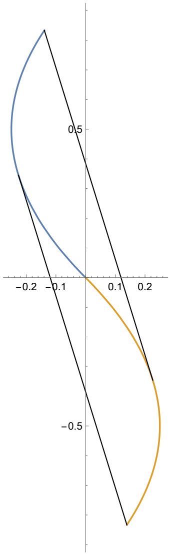
$K_0 = V_0 = 1$

$$
\begin{aligned}
\xi_D &= \left\{ \begin{array}{cc} 0.714 & 5 \\ 0.5 & 0.5 \end{array} \right\}, \\
\xi_{c_K} &= \left\{ \begin{array}{cc} 0.527 & 5 \\ 0.71 & 0.29 \end{array} \right\}, \\
\xi_{c_V} &= \left\{ \begin{array}{cc} 0.527 & 5 \\ 0.38 & 0.62 \end{array} \right\}.
\end{aligned}
$$

Sensitivity function for D-optimality



Elfving locus for c-optimality



The convex hull of

$$\left\{ \left(\frac{-V_0 x}{(K_0 + x)^2}, \frac{x}{K_0 + x} \right), x \in [0, bK] \right\}$$

and its reflection through the origin.

Michaelis-Menten extension

- y normally distributed with constant variance and mean:

$$E(y|x) = \frac{Vx}{K+x} + Fx \equiv \eta(x,\theta), x \in [0, bK], \theta^T = (K, V, F).$$

- Equivalent linear model (FIM):

$$f(x) = \left(\frac{\partial \eta}{\partial \theta}\right)_{(K_0, V_0, F_0)} = \left(\frac{-Vx}{(K+x)^2}, \frac{x}{(K+x)}, x\right).$$

- FIM: $M(\xi) = \sum_x \xi(x) f(x) F^T(x)$.

Results for $K_0 = V_0 = 1$:

$$
\begin{aligned}
\xi_{c_K} &= \left\{ \begin{array}{ccc} 0.31 & 2.37 & 5 \\ 0.55 & 0.33 & 0.12 \end{array} \right\}, \\
\xi_{c_V} &= \left\{ \begin{array}{ccc} 0.31 & 2.37 & 5 \\ 0.37 & 0.44 & 0.19 \end{array} \right\}, \\
\xi_{c_F} &= \left\{ \begin{array}{ccc} 0.31 & 2.37 & 5 \\ 0.28 & 0.43 & 0.29 \end{array} \right\}.
\end{aligned}
$$

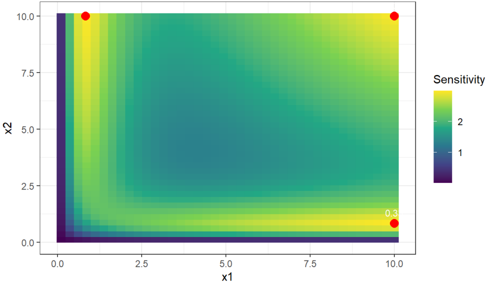
Two variables

y normally distributed with constant variance and mean:

$$E(y|x) = \frac{Vx_1x_2}{(K_1 + x_1)(K_2 + x_2)} \equiv \eta(x, \theta).$$

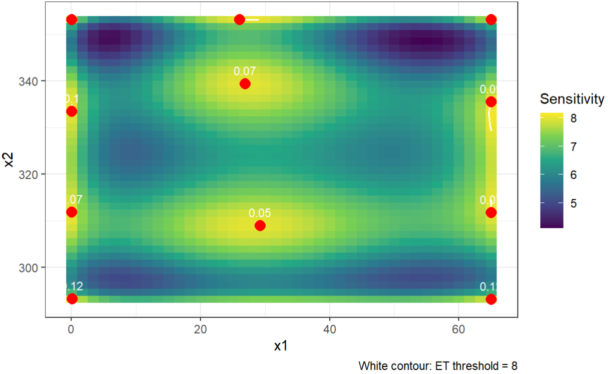
D-Optimal design:

x_1	x_2	Weight
0.83	10.00	1/3
10.00	0.83	1/3
10.00	10.00	1/3



White contour: ET threshold = 3

Heteroscedastic



D-optimal design of the Tait equation for 1-phenyldecane (8 parameters, 2D design space).

Summary

- General model: $h(y|x,\theta)$.
  FIM: $M(\xi,\theta)=\sum_{x\in\chi}\xi(x)\mathrm{E}_h\left[\left(\frac{\partial L(\theta)}{\partial\theta}\right)\left(\frac{\partial L(\theta)}{\partial\theta^T}\right)\right]=-\mathrm{E}_h\left[\left(\frac{\partial^2 L(\theta)}{\partial\theta^2}\right)\right]$.
- Normal, homoscedastic response:
  $M(\xi,\theta)=\sum_{x\in\chi}\xi(x)\frac{\partial\eta(x,\theta)}{\partial\theta}\frac{\partial\eta(x,\theta)}{\partial\theta^T}$.
- Linear model: $M(\xi,\theta)=\sum_{x\in\chi}\xi(x)f(x)f^T(x)$.
- Optimality criterion: $\Phi:\mathcal{M}\to[0,+\infty)$ to be minimized.
- Equivalence theorem: $\psi(x,\xi)=\partial\Phi[M(\xi),f(x)f^T(x)]=$
  $\nabla\Phi[M(\xi)][f(x)f^T(x)-M(\xi)]\geq 0,\ x\in\chi$.
- Analytical or algorithmic computation (based or not on the GET).

The optimal design is just a reference!

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